

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 7

5 Formal Models of Language (pjb48)

The Earley parser is an example of a top-down LR chart parser. This question requires you to design a *bottom-up RL chart parser* for parsing sentences with a context-free grammar.

charts for parsing (a) Describe the structure of the chart. [4 marks]

Answer: A chart is a dynamic programming table that contains edges [1 mark]. Each edge minimally records a unique id [1 mark]; a span [1 mark] (the portion of the input string which is consistent with edge); some information about the grammar component associated with that span [1 mark] (terminal or non-terminal). The candidate may also mention keeping note of the history but this isn't required until part e).

initialisation (b) Explain how the chart is initialised. [2 marks]

Answer: The chart will be initialised by the right-most word in the sentence [1 mark]. If there are n words the edge would look something like [1 mark]:
 $e_o \quad [n-1, n] \quad word_1$

parsing step (c) Describe the parsing algorithm using rule induction notation to describe new edges that can be added to the chart. [6 marks]

Answer: We can add the following new edges to the chart rules:

- $\frac{e_x [i, k] A}{e_z [i, k] C}$ where $C \rightarrow A$ exists in the grammar [2 marks]
- $\frac{e_x [i, k] A \quad e_y [k, j] B}{e_z [i, j] C}$ where $C \rightarrow A B$ exists in the grammar [2 marks]

Describe systematic method for scanning edges to be induced from [1 mark]. Also need to mention that once we have exhausted the edge inductions then a new word edge is added [1 mark]

parsing success criteria (d) Describe the properties of the chart when a parse is found. [2 marks]

Answer: We know we have found a parse when we find the following edge:

$e_n \quad [0, n] \quad S$ where 0-n spans the whole sentence [1 mark] and S is the root non-terminal [1 mark].

using edge history (e) Explain how to reconstruct a parse tree from the chart. [2 marks]

Answer: The edges will need to record some *history* [1 mark], that is, each edge needs to note the edges that made this edge possible. If this is done then once we find a parse we can trace the history back to uncover the rules that allowed the parse [1 mark]

parse selection (f) Explain how the algorithm could choose between parses if more than one is found. [2 marks]

Answer: You could score the partial parses as you build the chart with eg. probabilities from a PCFG [1 mark] in the case that there is more than one parse we can choose the one with the best score [1 mark]

parse efficiency

- (g) Does your approach result in a compact chart? If not, how could you reduce the number of edges in a chart without losing information about the possible parses? [2 marks]

Answer: You could pack the chart [1 mark]. This would require checking the grammar info and span before adding a new edge. If it exactly matches a previous edge then don't add the new one. Place the history information in a list [1 mark].
